SECTION 2: DATA COMPRESSION & RUN-LENGTH ENCODING [10 MARKS]

4. Lossy vs Lossless Comparison: (4 Marks)

Complete the comparison table by ticking [✓] or stating True/False:

Statement / Scenario	Lossy	Lossless
Decompressed file is 100% identical to the original uncompressed file.		
Permanently removes background sounds that human ears cannot perceive.		
Suitable for executable software program code and medical records.		
Audio format MP3 and image format JPEG.		

5. Run-Length Encoding (RLE) Problem Solving: (6 Marks)

a) A sequence of pixel colours is stored as: `W W W W W B B B B G G G G G G W W`

Write the RLE compressed string using (count, colour) pairs: (3 Marks)

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b) An RLE string is given as: `04 R 03 Y 05 B 02 G`

Decompress the string into its original sequence: (2 Marks)

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c) Under what condition does RLE fail to reduce file size? (1 Mark)

SECTION 3: ERROR DETECTION METHODS [10 MARKS]

6. Parity Block Matrix Error Location: (5 Marks)

A computer transmits data using **Even Parity**. The received parity block grid is shown below:

Byte	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Parity Bit
Byte 1	1	0	1	1	0	0	1	0
Byte 2	0	1	1	0	1	1	0	0
Byte 3	1	1	0	1	0	1	1	1
Byte 4	0	0	1	0	1	0	1	1
Parity Byte	0	0	1	1	0	0	1	0

a) Identify which byte contains the transmission error: _____

b) Identify which bit column flipped: _____